University College London, 2018-2019
Econ 0107 – Macroeconomics I – Ralph Luetticke & Franck Portier

Midterm Exam

This is a 2 hours exam. No documents allowed. 100 points in total, 50 points for each problem.

I – General Equilibrium with Incomplete Markets

Consider the following setup: There is a continuum of households that have a felicity function over consumption, c, such that

$$u(c) = -\lambda^{-1} \exp(-\lambda c),$$

with $\lambda > 0$. Households face uninsurable income risk where income is given by

$$y \overset{i.i.d.}{\sim} N(\mu, \sigma).$$

Households have a time-preference rate of $\delta > 0$ and try to maximize

$$U = E_0 \sum_{t=0}^{\infty} \left(\frac{1}{1+\delta} \right)^t u(c_t).$$

Households can self-insure (trading uncontingent claims amongst each other) against income risk using a risk-less asset a that bears interest r. There is no borrowing constraint except for the long-run transversality condition $\lim_{t\to\infty}(1+r)^{-t}a_t = 0$.

1 – Write the household's planning problem in form of a Bellman equation. Argue briefly, why the i.i.d. assumption for y allows to reduce the state space to a single dimension.

2 – Derive the Consumption Euler Equation and show that under the optimal policy the following must hold:

$$U = -\lambda^{-1} \frac{1+r}{r} \exp\{-\lambda[c_0]\}.$$

3 – Guess and verify that the policy function $a(x)$, where $x = a + y$, takes the form

$$a'(x) = x - rA_0$$

Hints: (i) Show that under the guess x follows a Gaussian random walk (with drift), (ii) then, make use of the FOC derived in question **2**, (iii) use the result $E \exp(y) = \exp(\mu + \sigma^2/2)$.

4 – Solve for A_0. Show that for $r = \delta$ and $\lambda \neq 0$ assets drift to infinity. Interpret the expression for A_0 economically.

5 – Define a "stationary equilibrium" for the above economy. Does it exist? If not how would you define a recursive equilibrium (with constant interest rate) here?

6 – Show that expression for the equilibrium interest rate(s) is approximately:

$$\left(\frac{1+r}{r} \right)^2 (\delta - r) = (\lambda\sigma)^2 /2 \tag{1}$$

7 – For below questions, intuitive explanation would suffice. How many equilibria are there? Can you welfare rank them in general? If not, why? In that case, can you welfare rank them "behind a veil of ignorance" when nobody has accumulated claims against somebody else? How would you do this (w/o actual calculations!)?

Consider a pure endowment economy, where the representative agent makes choices regarding consumption (C) and savings (B) solving the following problem:

$$\max_{\{C_t,B_t\}_{t=0}^{\infty}} \quad E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma}$$

$$\text{s.t.} \quad Y_t + B_{t-1} \geq C_t + Q_t B_t,$$

where Q_t is the price of a risk-free bond delivering one unit of consumption in period $t+1$. The endowment Y_t is assumed to follow the exogenous process:

$$\frac{Y_t}{Y_{t-1}} = \varepsilon_t,$$

where $\log \varepsilon_t \sim N(0, s^2)$.

1 – Derive the first order conditions of the agent's problem and the market clearing condition.

2 – Define a competitive equilibrium of this economy. Find an expression for the equilibrium value of Q_t that involves β, Y_t, Y_{t+1}, σ and the conditional expectation operator. (call this equation (1))

3 – What is the value of Q in a perfect foresight (deterministic) steady state?

4 – Solve equation (1) to find an exact expression for the bond price Q_t as a function of constants and/or variables that are observable at time t or earlier. (You may need to use the following relationship: if X follows a log-normal distribution and $x = \log X$, then $E(X) = Ee^x = e^{E(x)+\frac{1}{2}Var(x)}$).

5 – What is the average value of Q? Is it the same as the steady state value computed in question **3**? Why or why not?

6 – How does the variance of the endowment process (s^2) affect the bond price? Why?

7 – Find the equilibrium solution for the interest rate $i_t = -\log Q_t$.

8 – Suppose that neither you nor your computer could find the exact solution obtained in the question above. Log-linearizing equation (1) around the steady-state, and obtaining an approximate solution for the interest rate might be an easier task. How good is such an approximation? To answer that question, show first that the Euler equation of the household can be written

$$E_t \left[\beta Q^{-1} \exp\left\{-\widehat{q}_t - \sigma(\widehat{y}_{t+1} - \widehat{y}_t)\right\}\right] - 1 = 0 \qquad (\star)$$

where for a generic variable x_t we have used the definition $\widehat{x}_t = \log X_t - \log X$. Then linearize (– *i.e.* take a first order Taylor expansion) the equation ($\star$) and use it to obtain an approximate solution for the interest rate as a function of constants and/or variables which are observable at time t or earlier. Show how to obtain the expression $-\widehat{q}_t - \sigma E_t[\widehat{y}_{t+1} - \widehat{y}_t] = 0$. How does the interest rate depend on the variance of the endowment process according to this approximate relation?

9 – Assume that the standard deviation of the endowment process (denoted s) can take values in the interval $[0.01, 0.1]$. Comparing the expressions obtained in questions **7** and **8**, calculate the approximation error (in percentage points), first assuming that the agent is risk-neutral ($\sigma = 0$) and then assuming that the agent has a degree of risk-aversion $\sigma = 5$. Discuss